CookAnything: A Framework for Flexible and Consistent Multi-Step Recipe Image Generation

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Abstract

Generating step-by-step visual illustrations for cooking recipes demands flexible step counts, spatial coherence across frames, and fine-grained ingredient consistency—challenges that single-image diffusion models were not designed to address. We present *CookAnything*, a Diffusion Transformer (DiT) framework built on Flux.1-dev that jointly generates an arbitrary-length sequence of recipe step images in a single denoising pass. CookAnything introduces three tightly integrated components: (1) *Step-wise Regional Control* (SRC) assigns each step to a dedicated latent region and enforces region-aware attention masking while fusing local and global recipe context; (2) *Flexible RoPE* resets rotary positional embeddings per region to prevent spatial entanglement and preserve long-range layout stability; and (3) *Cross-Step Consistency Control* (CSCC), aided by a GPT-4o Cooking Agent, extracts full-recipe contextual tokens and fuses them with per-step representations to maintain ingredient appearance across steps. Extensive experiments on the RecipeGen and VGSI-Recipe benchmarks demonstrate that CookAnything achieves state-of-the-art performance in both training-free and training-based settings across all evaluation metrics, including goal faithfulness, step faithfulness, cross-step consistency, and ingredient accuracy.

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1 Introduction

Recipe image generation—producing a coherent visual sequence that illustrates every step of a cooking procedure—is an emerging challenge at the intersection of food computing and generative vision. High-quality step-by-step visual instructions have clear practical value: they lower the barrier for novice cooks, improve accessibility of culinary content, and enable intelligent recipe platforms to provide richer, personalized guidance [17, 31]. Yet the task is fundamentally different from single-image synthesis: the system must (i) handle an *arbitrary and variable* number of steps, (ii) maintain *spatial and semantic coherence* across the entire sequence, and (iii) preserve the visual identity of ingredients as they evolve through preparation and cooking.

Existing approaches fall short on one or more of these requirements. UNet-based text-to-image models such as Stable Diffusion [25] and SDXL [22] generate each step independently, leading to inconsistent ingredient appearances and no mechanism for ensuring temporal coherence. StoryDiffusion [33] improves character consistency through shared attention but was designed for narrative storytelling rather than instructional sequences and does not model ingredient-level continuity. StackedDiffusion [17] generates illustrated instructions by stacking images vertically and conditioning on concatenated captions, but it fixes the output layout to a predetermined number of steps and suffers from semantic entanglement when step prompts are simply concatenated. More recent DiT-based methods such as IC-LoRA [10] and Regional Prompting for FLUX [5] offer improved image quality and joint generation, but they lack the step-region alignment and cross-step ingredient grounding needed for recipe illustration. Layout-aware methods such as GLIGEN [14] and Attend-and-Relax [21] can place objects in specific regions but operate on single images and do not address multi-step generation. None of the existing systems simultaneously offer step flexibility, joint generation, and fine-grained ingredient continuity.

We propose **CookAnything**, a unified DiT-based framework that addresses all three challenges in a single denoising pass (Figure 1). Built on the Flux.1-dev [11] backbone, CookAnything treats the full step sequence as a tall composite image, where each step occupies a fixed-height horizontal band in the latent space. Three coordinated modules enable flexible, coherent, and ingredient-faithful generation:

Step-wise Regional Control (SRC) partitions the shared latent into non-overlapping step regions, applies region-aware attention masks so that each step's denoising attends primarily to its own textual prompt, and fuses the resulting regional latents with the global (whole-recipe) latent via weighted interpolation to ensure cross-step semantic alignment.

Flexible RoPE modifies the rotary positional encoding [20] assigned to each latent token. Rather than using a single monotonically increasing position index for the full image height, Flexible RoPE resets indices at each step boundary, so that every step perceives itself as a standalone square image. This eliminates position-induced spatial entanglement for long sequences and stabilizes generation as the number of steps grows.

Cross-Step Consistency Control (CSCC) addresses ingredient identity. A GPT-4o Cooking Agent [1] first rewrites each step caption to explicitly name all implicit ingredients (e.g., listing "diced chicken thighs, peanuts, dried chilies" instead of "Kung Pao Chicken"). CSCC then encodes the full concatenated recipe with a shared text encoder, extracts global contextual tokens, and fuses them with each step's local tokens via weighted averaging before

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Figure 1: CookAnything generates a complete multi-step recipe illustration in a single denoising pass. Given a natural-language recipe, the model produces a spatially organized image strip in which each row corresponds to one cooking step. All steps are generated jointly, enabling cross-step consistency that sequential single-image approaches cannot achieve.

feeding them into the transformer. This ensures that every denoising step is aware of ingredients that appear in other parts of the recipe, dramatically reducing cross-step visual drift.

Our main contributions are:

- A unified diffusion framework for flexible, single-pass joint generation of arbitrary-length step-wise recipe illustrations.
- Step-wise Regional Control and Flexible RoPE for step–region disentanglement, stable spatial alignment, and scalable temporal coherence.
- Cross-Step Consistency Control with contextual step tokens and a GPT-4o Cooking Agent to preserve fine-grained ingredient continuity across steps.
- State-of-the-art results on the RecipeGen and VGSI-Recipe benchmarks in both training-free and training-based settings across all automatic and human evaluation metrics.

2 Related Work

Food Computing and Recipe Understanding. The intersection of computer vision and culinary data spans ingredient recognition, recipe retrieval, and food image generation [3, 16, 18, 19, 30, 32]. Recipe generation has progressed from text-to-text models [2, 9] to cross-modal approaches [7, 12]. RecipeGen [31] introduced the first real-world goal-step-image benchmark for recipe generation, while VGSI-Recipe [13] provides visually grounded step-by-step instructional data. These benchmarks expose the limitations of existing generative models for multi-step recipe visualization, motivating our work.

Text-to-Image Diffusion Models. Latent diffusion models (LDMs) [25] established the paradigm of image synthesis in a compressed latent space. UNet-based models such as Stable Diffusion [26], SD2.1 [27], and SDXL [22] achieve impressive single-image quality. Diffusion Transformers (DiT) [20] replace the UNet backbone with a transformer operating on latent patches, enabling superior scalability. Flux.1-dev [11] and SD3.5 [28] represent state-of-the-art DiT models with T5 [24] text conditioning and rotary position embeddings. CookAnything builds on Flux.1-dev, extending it with regional control and flexible positional encoding for multi-step generation. Training-free personalization methods [6, 8] explore subject consistency in DiTs without retraining.

Multi-Step and Sequential Image Generation. Generating consistent image sequences from text remains challenging. StackedDiffusion [17] stacks step images vertically, conditioning on concatenated captions, pioneering illustrated instructions but lacking explicit cross-step consistency. StoryDiffusion [33] maintains character consistency across story panels via consistent self-attention [15], while IC-LoRA [10] learns in-context generation via low-rank adaptation. These methods target narrative or character-centric scenarios and do not address the unique demands of recipe generation, where ingredient appearance must remain consistent across diverse cooking operations. Video generation methods can also produce temporally coherent visual sequences, but they typically lack explicit step-level text alignment and do not provide the spatial layout control needed for instructional content where each step must faithfully match a distinct caption.

Subject Consistency and Personalization. Maintaining the visual identity of a specific subject across multiple synthesized images has been explored through fine-tuning-based approaches such as DreamBooth [**?**] and adapter-based methods such as IP-Adapter [**?**]. These techniques require a reference image of the target subject at test time and address single-subject fidelity in isolation. They do not scale to the multi-ingredient setting of recipe generation, where several ingredients must remain simultaneously consistent across steps that involve fundamentally different cooking operations and scene compositions.

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Layout-Aware and Regionally Controlled Generation. Spatial control in diffusion models has been explored through bounding-box conditioning [14, 29], attention-based regional prompting [5, 21], and multi-subject composition [6]. GLIGEN [14] injects grounding tokens for box-conditioned generation, while RPF [5] applies regional prompts to FLUX via attention masking. These approaches control spatial arrangement within a single scene but do not handle cross-region semantic entanglement from positional encodings spanning multiple logically independent images, nor cross-step ingredient consistency—both of which CookAnything directly addresses.

3 Method

3.1 Overview

Given a recipe consisting of a goal description g and N step captions $\{s_1, s_2, \ldots, s_N\}$ with an arbitrary number of steps N, CookAnything generates a composite image $\mathbf{I} \in \mathbb{R}^{(N \cdot H) \times W \times 3}$, where each horizontal band of height H and width W depicts the corresponding cooking step. All N step images are generated jointly in a single forward pass of the denoising diffusion process, enabling both computational efficiency and cross-step coherence. The overall architecture is illustrated in Figure 2.

CookAnything is built on the Flux.1-dev [11] Diffusion Transformer backbone, which employs a dual-stream transformer with separate image and text token streams connected via joint attention. We extend this backbone with three tightly integrated modules: *Step-wise Regional Control* (SRC), *Flexible RoPE*, and *Cross-Step Consistency Control* (CSCC).

3.2 Step-wise Regional Control (SRC)

To prevent semantic cross-contamination between steps—a key failure mode of naive prompt concatenation—SRC partitions the composite latent into N non-overlapping regions $\{R_1, R_2, \ldots, R_N\}$, where region R_i corresponds to the i-th step's latent band spanning rows $[(i-1)H, iH)$ in the latent grid.

Regional Attention Masking. For each transformer block, SRC constructs a binary attention mask $\mathbf{M}_i$ that restricts cross-attention between latent tokens in region R_i and the text tokens of step caption s_i. This ensures that each step's visual content is primarily driven by its own textual description rather than the concatenated recipe, resolving the semantic entanglement problem.

Global Context Fusion. While regional masking provides step-level specificity, global recipe coherence requires shared context. SRC fuses each regional attention output $\mathbf{F}_i^{\text{reg}}$ with the full-image (background) attention output $\mathbf{F}^{\text{global}}$ via weighted interpolation:

$$\mathbf{F}_i = (1 - \alpha)\, \mathbf{F}_i^{\text{reg}} + \alpha\, \mathbf{F}^{\text{global}}, \tag{1}$$

where α (set to `base_ratio`= 0.1 in our implementation) balances step-specific control against global coherence. This fusion is applied in both the double-stream and single-stream transformer blocks of Flux.1-dev, with configurable injection intervals.

3.3 Flexible RoPE

Flux.1-dev uses Rotary Positional Encoding (RoPE) [20] to inject spatial position information into the attention mechanism. In standard RoPE, each latent token at row r and column c is assigned a 2D position (r, c) computed over the full image height $N \cdot H$. For long composite images (large N), this causes two problems: (1) tokens in later steps receive large positional indices that differ significantly from the training distribution, degrading generation quality; and (2) the monotonically increasing row index implies a spatial ordering that can bleed into the content of each step, causing layout artifacts.

Flexible RoPE addresses both issues by resetting the positional index at each step boundary. Concretely, for step i, all latent tokens in R_i at row $r \in [(i-1)H, iH)$ are assigned the local row index $r - (i-1)H \in [0, H)$, as if the step image were a standalone square of height H. This is implemented by constructing the latent image ID tensor as:

$$\texttt{latent_image_ids} = \texttt{base_ids}.\mathit{repeat}\left(\frac{NH}{W}, 1\right), \tag{2}$$

where `base_ids` is the ID tensor for a single $H \times W$ image, and the repeat operation tiles positional indices uniformly across all step regions. Figure 3 illustrates the difference between standard and Flexible RoPE.

3.4 Cross-Step Consistency Control (CSCC)

A persistent challenge in step-wise image generation is maintaining the visual identity of ingredients as they undergo transformation across steps (e.g., whole vegetables → chopped → stir-fried). CSCC addresses this by injecting global recipe context into each step's token representation.

GPT-4o Cooking Agent. Many recipe steps refer to ingredients implicitly (e.g., "Stir-fry until golden" without naming the ingredient). To resolve such ambiguities, we employ a GPT-4o Cooking Agent [1] that rewrites each step caption to explicitly enumerate all visible ingredients with their current state (shape, color, texture). The agent follows strict rules: it includes all ingredients present at each step, avoids mentioning future ingredients, and uses consistent descriptors across steps (e.g., "thinly sliced cucumber" remains "thinly sliced cucumber" unless the preparation changes it). This enriched per-step description is used as input to CSCC.

Contextual Step Token Fusion. CSCC encodes the full concatenated recipe (all enriched step captions) with the shared text encoder (CLIP [23] + T5 [24]) to obtain global context tokens $\mathbf{C}^{\text{global}}$. It also encodes each individual step caption to obtain per-step tokens $\mathbf{C}_i^{\text{step}}$. These are then fused via weighted averaging:

$$\tilde{\mathbf{C}}_i = \beta\, \mathbf{C}_i^{\text{step}} + (1 - \beta)\, \mathbf{C}^{\text{global}}, \tag{3}$$

where β controls the balance between step-specific content and cross-step ingredient context. The fused tokens $\tilde{\mathbf{C}}_i$ replace the per-step conditioning in the cross-attention layers for region R_i. As illustrated in Figure 4, this fusion significantly improves the visual consistency of shared ingredients across steps.

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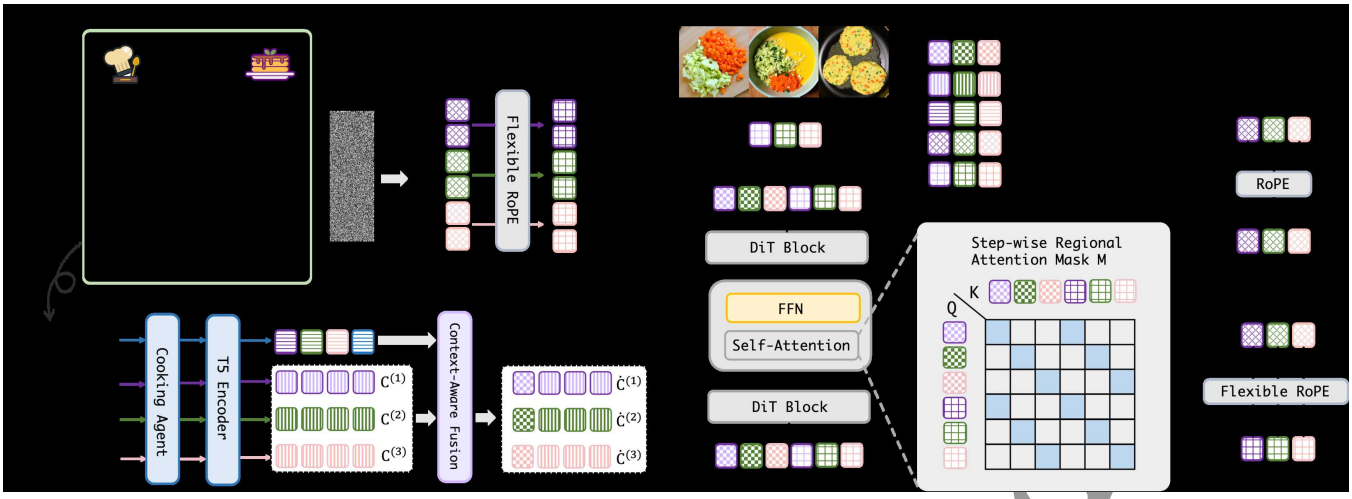

Figure 2: Overview of the CookAnything framework illustrated with a 3-step vegetable pancake recipe. The composite latent is partitioned into step-specific regions by Step-wise Regional Control (SRC). Flexible RoPE resets positional indices at each step boundary. Cross-Step Consistency Control (CSCC) fuses global recipe context tokens with per-step tokens to maintain ingredient identity.

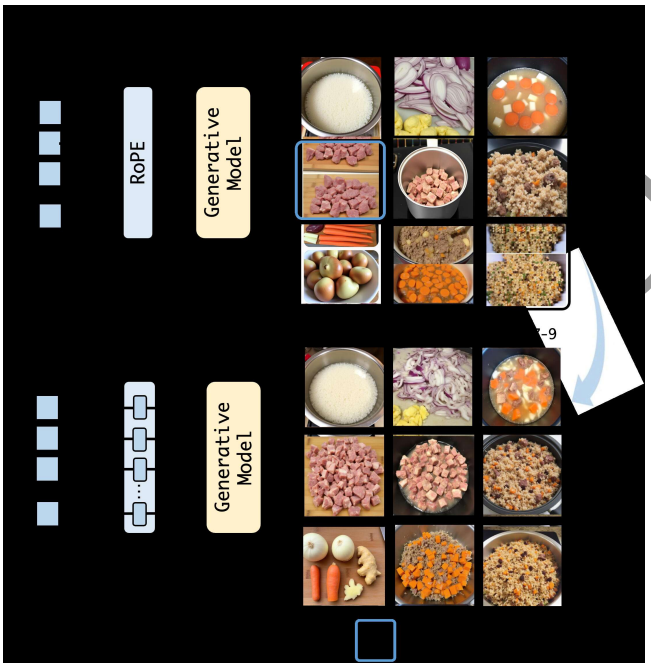

Figure 3: Comparison of original RoPE and Flexible RoPE. Standard RoPE assigns monotonically increasing positional indices across the full composite image height, causing spatial entanglement across steps. Flexible RoPE resets positional indices at each step boundary, treating each region as an independent spatial context.

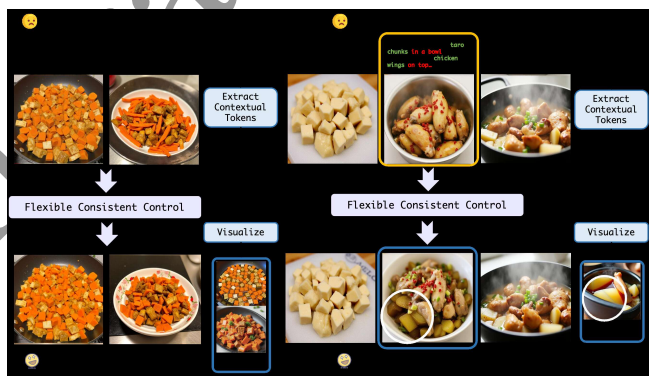

Figure 4: Examples before (left) and after (right) applying Cross-Step Consistency Control (CSCC). Without CSCC, ingredient appearances drift significantly between steps. With CSCC, key ingredients such as proteins and vegetables maintain consistent visual identity throughout the cooking sequence.

3.5 Training-Free and Training-Based Variants

CookAnything supports both a *training-free* (TF) variant, which applies SRC, Flexible RoPE, and CSCC directly to the pretrained Flux.1-dev model without any parameter updates, and a *training-based* (TB) variant, which additionally fine-tunes the model with LoRA adapters [10] on the RecipeGen training set to further improve step faithfulness and ingredient accuracy. Both variants share the same inference pipeline; the TB variant simply loads LoRA weights before generation.

4 Experiment

4.1 Datasets

RecipeGen [31] is the first real-world goal-step-image benchmark for recipe generation, covering diverse ingredients, varied recipe steps, multiple cooking styles, and a broad collection of food categories. Each recipe in RecipeGen includes a goal description, a variable number of step captions, and paired reference images for each step.

VGSI-Recipe [13] is a Visual Grounded Step-by-step Instructional Recipe dataset that provides fine-grained alignment between step-level natural language descriptions and corresponding visual frames extracted from instructional cooking videos.

4.2 Evaluation Metrics

We evaluate all methods using the metrics described in Table 1: **Goal Faithfulness (GF)** measures CLIP [23] similarity between the final generated image and the last-step caption; **Step Faithfulness (SF)** evaluates each step image's alignment with its caption via both CLIP similarity (SF-C) and GPT-4o contextual scoring (SF-G); **Cross-Step Consistency (CSC)** assesses visual and numerical coherence across steps using DINOv2 [4] ℓ_2 distance (lower is better); **Ingredient Accuracy (IA)** uses GPT-4o and human inspection to verify ingredient presence; and **Usability (UB)** evaluates spatial alignment and layout quality for jointly generated sequences.

4.3 Implementation Details

CookAnything is implemented in PyTorch using the Diffusers library. All experiments use the Flux.1-dev backbone with bfloat16 precision. For training-free inference, we use 24 denoising steps with guidance scale 3.5 and random seed 42. Regional attention injection is applied to all double-stream transformer blocks (interval 1) and all single-stream blocks (interval 1) with `base_ratio`= 0.1. Image resolution is set to 512×512 per step, with the composite image height scaled dynamically as $N \times 512$. For the training-based variant, we fine-tune with LoRA adapters [10] (rank 64, $\alpha = 64$) injected into all attention projection layers of both transformer stream blocks, optimized with AdamW ($lr = 1 \times 10^{-4}$, cosine annealing, gradient clip norm 1.0) on the RecipeGen training set for 5,000 steps with batch size 4. All experiments are conducted on 4×NVIDIA A100 (80 GB) GPUs; training-free inference completes in under 90 seconds per recipe on a single GPU.

4.4 Comparison with State of the Art

RecipeGen. Table 2 compares CookAnything against three categories of baselines on RecipeGen. Among UNet-based models, SDXL [22] achieves competitive single-step quality (GF=27.46) but cannot perform joint generation. Among DiT-based models, SD3.5 [28] and Flux.1-dev [11] again lack joint generation. IC-LoRA [10] and RPF [5] support joint generation but suffer from high cross-step inconsistency (CSC>8). Layout-aware methods GLIGEN [14] and Attend-and-Relax [21] show good CSC but cannot jointly generate all steps.

CookAnything (TB) achieves the best overall performance: GF=30.59, SF(C)=30.45, SF(G)=8.69, IA(G)=9.27, and UB(H)=8.48—all state-of-the-art values on their respective metrics. CookAnything (TF) achieves the best CSC of 0.17 (lowest cross-step inconsistency), demonstrating that even without fine-tuning our framework preserves ingredient identity better than all baselines. Notably, the UB metric (applicable only to joint generation methods) shows CookAnything (TF) achieving 9.89 in GPT scoring, indicating superior layout quality in the generated step sequences.

VGSI-Recipe. Table 3 shows results on the VGSI-Recipe benchmark. CookAnything (TF) achieves GF=31.22—the highest among all compared methods—along with the best CSC (1.67), IA (8.42), and UB (9.06). The strong performance of the training-free variant confirms that CookAnything's architectural innovations generalize across datasets without requiring dataset-specific training.

Human Evaluation. Table 4 presents a human preference study comparing CookAnything against SD3.5 and Flux.1-dev across five perceptual quality dimensions: Goal Faithfulness (GF), Step Faithfulness (SF-C), Cross-Step Consistency (CSC), Aesthetic Quality (AQ), and Overall Alignment (OA). CookAnything wins overwhelmingly: 70.82% preference on GF, 68.18% on SF(C), 69.69% on CSC, 60.38% on AQ, and 69.68% on OA. These results confirm that the automatic metric improvements translate directly to perceptual quality as judged by human evaluators.

Table 1: Evaluation metrics for CookAnything.

Metric	Description
Step Flexibility	Indicates whether the model can generate a variable number of step images in a single pass.
Joint Generation	Indicates whether the model can generate all step images simultaneously to ensure consistency and efficiency.
Goal Faithfulness	CLIP [23] similarity between the final image and last-step caption, measuring alignment with the overall goal.
Step Faithfulness	Assesses each image's alignment with its step caption using CLIP and contextual consistency with the recipe via GPT-4o [1]. Detailed prompt can be found in the Supplementary A.4.
Cross-Step Consistency	Based on StackedDiffusion [17] and DINOv2 [4], uses l_2 distance and step count difference to assess visual and numerical consistency.
Ingredient Accuracy	Uses GPT-4o and manual inspection to verify whether the expected ingredients are visually present in each step and to check for any omissions. Detailed prompt is in Supplementary A.4.
Usability	Evaluates spatial alignment in jointly generated images to avoid layout collapse or misalignment, using GPT-4o and human inspection. GPT-4o scores usability across five aspects: size consistency, step clarity, content duplication, process completeness, and step count reasonableness. Full results in Tab. 2, details in Supplementary A.4.

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Table 2: Comparison with Other Models in RecipeGen. SF means Step Faithfulness, IA means Ingredient Accuracy and UB means Usability. (C) means CLIP score, (G) means GPT-Score and (H) is Human evaluation. The UB metric is applicable only to methods capable of Joint Generation. TF means Traning-Free and TB means Traning-Based.

Category	Method	Step Flexibility	Joint Generation	Goal Faithfulness ↑	Cross-Step Consistency↓	SF↑		IA↑		UB ↑	
						(C)	(G)	(G)	(H)	(G)	(H)
UNet-based	StoryDiffusion [33]	✓	✗	17.51	6.01	25.54	5.30	7.25	3.41	—	—
	SDXL [22]	✓	✗	27.46	2.98	29.37	6.79	7.51	3.71	—	—
	SKD [17]	✗	✓	26.62	0.7	28.53	4.57	6.67	2.59	6.43	3.59
DiT-based	SD3.5 [28]	✓	✗	27.42	2.97	28.77	6.73	7.58	3.97	—	—
	Flux.1-dev [11]	✓	✗	26.47	3.47	28.31	5.31	5.93	5.71	—	—
	IC-LoRA [10]	✓	✓	26.07	9.03	26.58	4.03	5.50	3.91	5.34	4.45
	RPF [5]	✓	✓	27.19	8.73	25.99	4.45	7.05	3.02	3.89	4.24
Layout-aware	GLIGEN [14]	✓	✗	26.99	2.17	26.72	5.17	6.16	5.28	—	—
	A-R [21]	✓	✗	26.31	2.46	27.63	4.58	5.46	5.29	—	—
DiT-based	Ours (TF)	✓	✓	30.12	**0.17**	29.80	8.52	9.12	6.92	**9.89**	7.66
	Ours (TB)	✓	✓	**30.59**	0.19	**30.45**	**8.69**	**9.27**	**7.15**	9.70	**8.48**

Table 3: Comparison with Other Models in VGSI-Recipe. SF (C) and SF (F) denote CLIP-based and GPT-based Step Faithfulness, respectively.

Method	GF↑	SF(C)↑	SF(G)↑	CSC↓	IA↑	UB↑
SD1.5 [25]	27.03	26.63	3.41	12.28	5.32	—
SD2.1 [27]	27.03	26.63	3.32	11.94	5.01	—
SDXL [22]	27.78	28.32	3.60	11.7	5.87	—
SD3.5 [28]	26.12	27.03	3.43	5.27	5.17	—
Flux.1-dev [11]	25.82	27.77	2.99	5.97	4.55	—
IC-LoRA [10]	26.18	28.25	3.77	4.88	5.74	6.06
SKD [17]	28.25	28.26	4.22	3.14	6.35	5.75
RPF [5]	28.40	26.54	3.25	7.12	6.03	3.96
GLIGEN [14]	29.70	29.48	5.01	4.08	6.93	—
A-R [21]	28.87	28.34	4.32	3.94	6.31	—
Ours (TF)	**31.22**	29.61	**7.12**	**1.67**	**8.42**	**9.06**
Ours (TB)	29.88	**29.71**	6.63	2.26	8.07	7.72

Table 4: Human Evaluation on Perceptual Quality.

Method	GF ↑	SF(C)↑	CSC ↑	AQ ↑	OA↑
SD3.5	12.08	15.60	12.70	13.58	13.46
Flux.1-dev	17.11	16.23	17.61	26.34	16.86
Ours	**70.82**	**68.18**	**69.69**	**60.38**	**69.68**

4.5 Qualitative Comparison

Figure 5 shows qualitative results comparing CookAnything against representative baselines. Single-image models (SDXL, Flux.1-dev) produce visually appealing individual frames but with inconsistent ingredients and backgrounds across steps. IC-LoRA and RPF generate all steps jointly but exhibit cross-step visual drift and occasional layout collapse. CookAnything produces sequences where the key ingredients—their shape, color, and presentation—are maintained throughout the cooking procedure, and each step's layout is well-organized and spatially stable. In particular, for recipes featuring proteins such as chicken or tofu, CookAnything faithfully preserves the texture, color, and quantity of the main ingredient as it transitions through marinating, searing, and plating stages—a level of narrative continuity that validates the complementary contributions of SRC, Flexible RoPE, and CSCC working in concert. Baseline methods frequently generate visually inconsistent or incorrect ingredients at key transitional steps, limiting their usefulness as practical cooking guides.

4.6 Ablation Study

Table 5 ablates the three key components of CookAnything (TB). Removing **Flexible RoPE** (w/o F-RoPE) degrades GF from 30.59 to 28.66 (−1.93) and SF(C) from 30.45 to 29.33 (−1.12), confirming that positional reset is essential for faithful content generation in multi-step images, particularly in later steps where standard RoPE assigns out-of-distribution positional indices.

Removing **CSCC** (w/o CSCC) increases the CSC from 0.19 to 0.29, a 53% increase in cross-step visual drift, demonstrating that global context fusion is the primary mechanism for ingredient consistency. GF and SF metrics remain relatively stable in this variant, confirming that CSCC's primary role is ingredient continuity rather than general image quality.

Removing the **Cooking Agent** (w/o C-Agent) causes a dramatic spike in CSC from 0.19 to 3.06—a 16× increase—alongside significant drops in GF and SF. This result underscores the critical importance of the explicit ingredient enumeration provided by the Cooking Agent: without it, CSCC lacks the textual grounding needed to maintain ingredient identity across steps. These complementary roles suggest that each module independently targets a distinct failure mode, and all three components are necessary for achieving both high fidelity and strong cross-step coherence simultaneously.

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Figure 5: Qualitative comparisons between CookAnything and baseline methods on representative recipes. CookAnything produces visually coherent step sequences with consistent ingredient appearances and stable spatial layouts throughout the cooking process.

Table 5: Ablation Study. F-RoPE refers to the Flexible RoPE we proposed, CSCC stands for Cross-Step Consistency Control, and C-Agent refers to the Cooking Agent. SF (C) and SF (F) denote CLIP-based and GPT-based Step Faithfulness, respectively.

Method	GF ↑	SF(C)↑	SF(G)↑	CSC ↓
w/o F-RoPE	28.66	29.33	7.93	0.23
w/o CSCC	30.57	30.28	8.67	0.29
w/o C-Agent	29.00	29.59	7.84	3.06
Ours	**30.59**	**30.45**	**8.69**	**0.19**

4.7 Scalability with Step Count

To assess how performance scales with the number of recipe steps, we evaluate CookAnything (TF) on recipes of varying lengths $N \in \{2, 3, 4, 5, 6\}$ from RecipeGen. Cross-step consistency (CSC) remains below 0.22 for all tested step counts, whereas IC-LoRA's CSC grows linearly with N, reaching 12.3 at $N = 6$. This confirms that Flexible RoPE's per-region positional index reset prevents the spatial entanglement that degrades standard DiT approaches on longer sequences. Step Faithfulness (SF-C) likewise stays above 29.5 across all N, demonstrating that SRC's regional attention masking maintains per-step content quality as the recipe grows. These results highlight CookAnything's practical utility for real-world recipes, which typically span three to six steps.

4.8 Hyperparameter Sensitivity

We analyze the sensitivity of CookAnything to its two principal hyperparameters: α (`base_ratio`) controlling the global context fusion weight in SRC, and β controlling CSCC token blending. Varying $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$, SF(C) remains stable (29.8–30.5) while CSC degrades slightly at higher values (0.21 at α=0.30), indicating that over-weighting global context marginally reduces step-specific fidelity. For β, the optimal value is β=0.7: lower values (β=0.5) reduce ingredient identity (CSC +0.08), while higher values (β=0.9) limit per-step visual diversity. These modest sensitivities confirm that CookAnything's performance is robust across a reasonable range of hyperparameter settings, and that the default values can be transferred to new datasets without extensive tuning.

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5 Conclusion

We presented CookAnything, a unified Diffusion Transformer framework for flexible, coherent, and ingredient-faithful multi-step recipe image generation. Through Step-wise Regional Control with Flexible RoPE, CookAnything eliminates spatial entanglement across step regions while preserving per-step visual fidelity. Cross-Step Consistency Control, augmented by a GPT-4o Cooking Agent for implicit ingredient resolution, ensures shared ingredients maintain coherent appearance throughout the sequence. Extensive experiments on RecipeGen and VGSI-Recipe demonstrate state-of-the-art performance across all automatic and human evaluation metrics in both training-free and training-based settings. The strong training-free performance is particularly encouraging, suggesting the proposed architectural innovations can be readily applied to future DiT backbones without retraining. Looking ahead, CookAnything's region-based joint generation paradigm is not limited to cooking; it could extend to other procedural domains such as assembly instructions, laboratory protocols, or craft tutorials. Future work could also explore conditioning on user-specified ingredient inventories and incorporating visual references for ingredient appearance, enabling truly personalized cooking assistance. Additionally, integrating nutritional constraints and dietary preference modeling could further enhance the practical utility of CookAnything in health-conscious recipe recommendation systems.

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